# Test Task: Research Investigation of Trust-Region Projections as a Replacement for PPO Clipping

The goal of this task is to conduct a small research investigation inspired by **TROLL**, which proposes replacing PPO-style clipping with a differentiable trust-region projection.

PPO clipping is widely used because it is simple and effective, but it is only an indirect way to limit policy updates. The TROLL paper argues that a more explicit trust-region projection can provide better controlled updates by projecting the current policy distribution back into a KL ball around the old policy.

In the original paper, this idea is studied for language-model training. In this task, you will investigate a smaller and more controlled version in standard discrete-action RL. The main goal is not to reproduce the paper's large-scale experiments, but to understand whether the same trust-region idea is useful when applied to PPO on ordinary environments.

### References

- Main paper: TROLL: Trust Regions improve RL for LLMs.
- Suggested implementations reference: CleanRL, EnvPool, Gymnasium.

### Research Objective

Investigate whether a TROLL-style trust-region projection can improve or change the behavior of PPO on small or medium-scale discrete-action environments.

A good submission should answer a research question of the following form:

> *Can an explicit KL trust-region projection replace PPO clipping in discrete-action RL, and under what conditions does it improve stability, sample efficiency, or robustness?*

You may refine this question in your own way. For example, you may focus on sensitivity to learning rate, number of PPO epochs, policy lag, KL bound, projection frequency, etc.

### Task

Design and run a small experiment that helps understand whether a trust-region projection can replace or improve PPO clipping in discrete-action RL.

A recommended starting point is to modify a CleanRL PPO implementation and run it with EnvPool on several discrete-action environments. Suitable environments include Atari-style environments. A compact and carefully analyzed experiment is preferred over a large sweep.

Your solution should include:

- a clear research question or hypothesis;
- an implementation of a TROLL-style trust-region projection for discrete action distributions;
- an experimental way to stress clipping, such as increasing learning rate, increasing PPO epochs, increasing batch size, or introducing stale rollout data;
- plots or tables comparing learning curves, final returns, and stability metrics;
- a discussion of when the projection helps, when it does not help, and whether it is worth the added complexity.

The environment, architecture, and exact protocol are up to you. However, you should keep the setup small enough that someone else can run it end-to-end.

**Expected Deliverables**

Please submit:

- source code or a notebook that can be run end-to-end;
- a short report, approximately 3–5 pages, including plots or tables with clear captions;
- a description of the exact environment, hyperparameters, and seeds used;
- a comparison against standard PPO clipping;
- a final recommendation explaining whether this projection-based update should be used in RL.

The outcomes should be written as a small research report. It should explain what you tried, why you chose this setup, what you observed, and what conclusions can or cannot be drawn from the experiment.